

Does Geography Predict Income?

Regional and Urban/Rural Determinants of County Median Household Income

Does geography predict county income?

Urban counties earn ~\$10,020 more than rural counties regardless of region ($p < 0.001$).

Model: $\text{Income} = \beta_0 + \beta_1(\text{Northeast}) + \beta_2(\text{South}) + \beta_3(\text{West}) + \beta_4(\text{Urban}) + \beta_5(\text{NE} \times \text{Urban}) + \beta_6(\text{South} \times \text{Urban}) + \beta_7(\text{West} \times \text{Urban}) + \varepsilon$ | Baseline: Rural Midwest | $R^2 = 0.304$

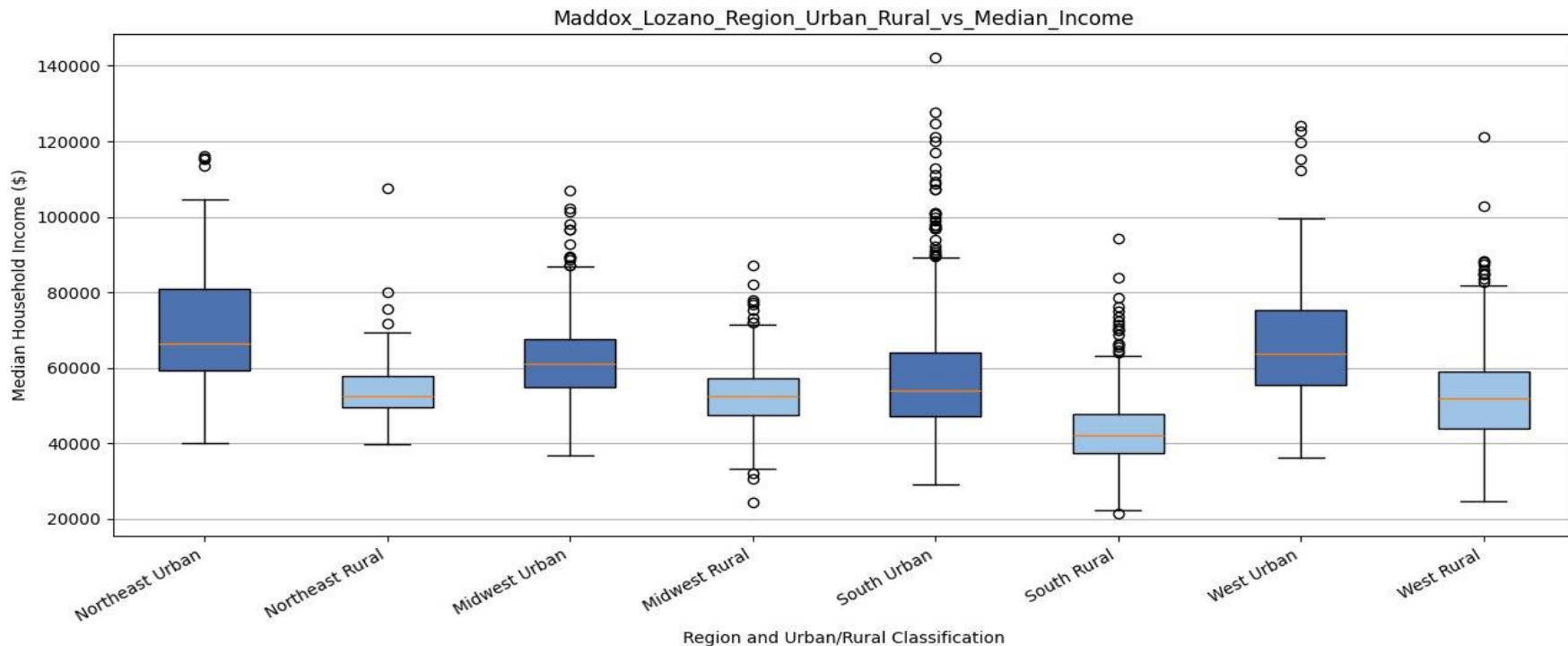


Figure 1. Median household income by census region and urban/rural classification, 3,140 U.S. counties.

What did I find, and does the model hold up?

Urban/rural classification predicts income more than region alone, but OLS assumptions are violated and robust errors were applied.

- Being in an urban county raises income more than being in any particular region
- Rural counties in the South have the lowest incomes of any group in the country
- The income boost from being urban is largest in the Northeast and smallest in the West
- The model had assumption violations. A correction was applied so the results are still reliable

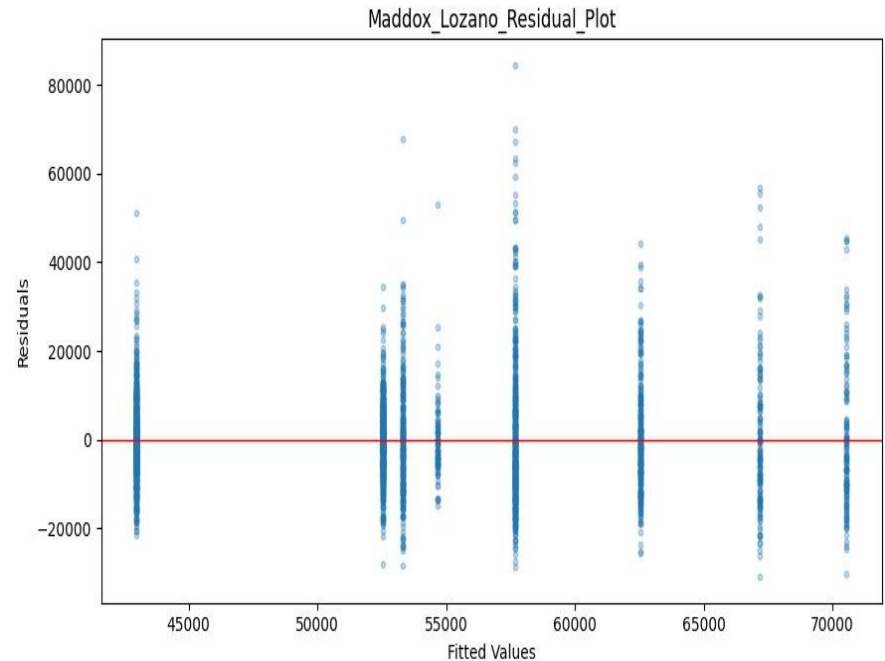


Figure 2. Residual plot: OLS assumption diagnostic.